CIA 01. Cryptography and Network Security.

CSA5111.

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11. PT: MEETATNOON. shift key : 7. Caesar Cipher.

C=(P+7) %. 26.

Encryption

M E E T . A T N O O N

12 4 4 19 0 19 13 14 14 13

(12+7) %. 26 = 19.

(19+7) %. 26 = 0

(13+7) %. 26 = 20

(4+7) %. 26 = 11

(0+7) %. 26 = 7.

(14+7) %. 26 = 21.

Ciphertext = T L L A H A U V V U

decryption.

T L L A H A U V V U

19 11 11 0 7 0 20 21 21 20

(19-7) %. 26 = 12

(11-7) %. 26 = 4

(7-7) %. 26 = 0

(20-7) %. 26 = 13

(21-7) %. 26 = 14

Vigenère cipher

Key: SUNSUNSUNS.

(12+18) %. 26 = 4.

(4+13) %. 26 = 17

(0+20) %. 26 = 20

(13+18) %. 26 = 5

(14+13) %. 26 = 1

C = P+K. %. 26

(4+20) %. 26 = 24.

(19+18) %. 26 = 11

(19+13) %. 26 = 6.

(14+20) %. 26 = 8.

(13+18) %. 26 = 5

E V R L U G F I B F -> Cipher text.

Comparision

	Caesar	Vigenere
Key space:	26 shifts	26^m
Repeated Patters.	Same letter Give same. CT.	same letter gives different. CT
Security.	Weak.	Strong.
Cryptanalisis	Easily broken with brute force.	broken with Kasiski and - frequency Analysis

Vigenere cipher is Recomended because of large Key Space.

it is not suitable for modern Security. because. both has small Key Spaces., Some time PT can produce Same CT; it does not detect data modificatio and both are vulnerable to cryptanalisis

2. Plain text : MATH. Key 3 2
5 7.

$C = KP \% 26$.

MATH → 12 0 19 7.

$$\begin{bmatrix} M & T \\ A & H \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \end{bmatrix} \begin{bmatrix} 19 \\ 7 \end{bmatrix}$$

$$= \left[\begin{array}{c|c} 12 & 19 \\ \hline 0 & 7 \end{array}\right] \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \% 26 = \begin{bmatrix} 36 & 71 \\ 60 & 144 \end{bmatrix} \% 26$$

$$= \begin{bmatrix} 10 & 19 \\ 8 & 14 \end{bmatrix} = \begin{bmatrix} K & T \\ I & O \end{bmatrix} \rightarrow CT.$$

$\det(k) = (3 \times 7) + (2 \times 5) = 21 - 10 = 11.$

$11^{-1} \% 26 = 11 \times 19 \% 26 = 1$

$11^{-1} = 19.$

$k^{-1} = \det(k)^{-1} \times \text{adj}(k) \% 26.$

$\text{adj}(k) = \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} =$

$k^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \% 26.$

$k^{-1} = \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \% 26.$

$$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

decryption

$$P = K^{-1} C \mod 26$$

$$= \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 10 & 19 \\ 8 & 14 \end{bmatrix} \mod 26$$

$$= \begin{bmatrix} 142 & 253 \\ 130 & 241 \end{bmatrix} \mod 26$$

$$= \begin{bmatrix} 12 & 19 \\ 0 & 7 \end{bmatrix} = \begin{bmatrix} M & T \\ A & H \end{bmatrix}$$

Alternate key

$$k = \begin{matrix} 2 & 3 \\ 4 & 5 \end{matrix} \qquad \det(k) = 2 \times 5 - 3 \times 4$$

$$= -2.$$

$$GCD(-2, 26) = 2.$$

$\therefore$ it is not Acceptable

<u>weakness of hill cipher.</u>

- -> it is a linear Encryption technique.
- -> Attacker can use known plain text to find key.
- -> does not provide data Intergrity
- -> Vulnerable to plain text known Attacks

3.

AES-256 GCM.	RSA.
One shared key	Public and private key.
Same Encrypts and decrypts.	Public key Encrypts. Private key decrypts
Both party hold the shared key	Private key remains secret.
Very fast and suitable for large files such as 1 GB MRI Archieves.	Slower and only for small data such as Encryption key.

Comparison table.	AES-256-GCM.	RSA.
Type.	Symmetric.	Asymmetric
Speed.	Fast	slow
file size.	large	small
Keys	shared secret.	public private pair.
usage	Data Encryption	Key Encryption

b) Hybrid Encryption of AES & RSA.

Sending

- Verify digital certificate and key.
- Generate 256 bit data Encryption key
- Generate unique 96 bit Nonce
- Encrypt MRI file using AES with AAD
- Encrypt DEK with RSA public key.
- Send CT, DEK, nonce, Tag, AAD to Server

Recieving,

- Get Encrypted Package.
- USE RSA to recover DEK.
- USE DEK and Nonce to decrypt the AAD
- Verify Tag.
- Recover MRI Data.

Key rotation

-> Generate new DEK for Each file.

-> Rotate the KEK.

-> Rewrap existing DEK using the new KEK.

-> Retire old key after use.

-> rotate compromised key during Incidents

4. RSA Algorithm

Encryption is done by computing algebraic notations with. Given p, q and calculating the totient, Euclised formula, it uses a key pair system to Encrypt the data. it uses Extended Euclidean Algorithm

for Given Value · p = 11 q = 19. public Exponent · e = 7 · d

formula.

$$\phi(n) = \frac{(p-1)\times(q-1)}{p-1} = (19-1)(11-1) = 18 \times 10$$

$$= 180.$$

we know $\phi(n) = 180$.

verify the possibility of RSA.

$$e \times d \equiv 1 \pmod{180} = 1$$

$$7 \times d \equiv 1 \pmod{180}$$

$$7 \times 26$$

$$d = 26.$$

Characteristics of RSA.

→ it is a public private key Encryption

→ public key is used to Encrypt the da

→ Private key is used to decrypt

→ Private key is kept secreat.

PLAYFAIR CIPHER.

KEY: CHRISTO

it is much slower for files. So it is ~~slow~~ ~~only~~ only used for key Encryption

Properties

-> it is slow for file Encryption
-> it is a Symetric type Encryption
-> Small file size.
-> it uses public private pair.
-> used for password and key Encryption

Usage

-> RSA - OAEP is used for Encryption of data.
-> RSA - PSS is used for digital Signature

Vigenère Cipher:

MEET AT NOON -> PT KEY = SUN

len(PT) = 10 ∴ KEY = SUNSUNSUNS

Encryption

PT -> 12 4 4 19 0 19 13 14 14 13.

Key -> 18 20 13 18 20 ~~2 0~~ 13 18 20 13 18.

12+18 %.26 = 30 %.26. = 04. = E

4+20 %.26 = 24 %.26 = 24 = Y.

4+13 %.26 = 17 %.26. = 17 = R

19+18 %.26 = 37 %.26. = 11 = L

0+20 %.26 = 20 %.26. = 20 = U

19+13 %.26 = 32 %.26. = 6 = G

13+18 %.26 = 31 %.26 = 5 = F

14+20 %.26 = 34 %.26. = 8 = I

14+13 %.26 = 27 %.26. = 1 = B.

13+18 %.26 = 31 %.26 = 5. = F

CT = E Y R L U G F I B F

decryption

CT = 4 24 17 11 20 6 5 8 1 5

Key 18 20 13 18 20 13 18 20 13 18.

4-18 %.26 = 12

24-20 %.26 = 4

17-13 %.26 = 4

11-18 %.26 = 19

20-20 %.26 = 0

6-13 %.26 = 19

5-18 %.26 = 13

8-20 %.26 = 14

1-13 %.26 = 14

5-18 %.26 = 13

PT = MEET AT NOON.

Encryption and decryption

5. Plain text: meet at noon. Shift 7.

Caesar cipher.

PT → Number ~~MEET~~ MEET AT NOON.

12 4 4 19 0 19 13 14 14 13.

Encryption $= (P + k) \bmod 26$.

12+7 = 19 Y. 26 = 19
4+7 = 11 Y. 26 = 11
4+7 = 11 Y. 26 = 11
19+7 = 26 Y. 26 = 0
0+7 = 7 Y. 26 = 7
19+7 = 26 Y. 26 = 0
13+7 = 20 Y. 26 = 20
14+7 = 21 Y. 26 = 21
14+7 = 21 Y. 26 = 21
13+7 = 20 Y. 26 = 20.

= 19 11 11 0 7 0 20 21 21 20

T L L A H A U V V U

decryption.

$= (C - K) \bmod 26$

= 19-7 = 12 Y. 26 = 12
= 11-7 = 4 Y. 26 = 4
= 11-7 = 4 Y. 26 = 4
= 0-7 = -7 Y. 26 = 19
= 7-7 = 0 Y. 26 = 0
= 0-7 = -7 Y. 26 = 19
= 20-7 = 13 Y. 26 = 13
= 21-7 = 14 Y. 26 = 14
= 21-7 = 14 Y. 26 = 14
= 20-7 = 13 Y. 26 = 13

PT = MEET AT NOON

PLAYFAIR CIPHER.

KEY: CHRISTO

PT: OPHER.

5×5 = 25 make a 25 Grid box.

C	H	R	I/J	S
T	O	A	B	D
E	F	G	K	L
M	N	P	Q	U
V	W	X	Y	Z

Rules:-

→ if both on same
- Column. take value to the imediate down
→ if same row take value from imediate right

make pairs ~~CH, RI, SO~~ X. OP, HE, RX.

NA GF X I

6 b) AES-256 Encryption

AES Encryption is used in most of the place because it is much more secure and Not vulnerable to basic cryptanalis, it has much larger Key Space; it is a Symetteric cypt cipher.

Components of AES-256-GCM.

nonce: 96 bit : Plain text.

Key. prime p.

AES-256 produces a fixed length output of 256 bits & And the cipher text is calculated using 96 bit nonce and mod p.

One Key is shared between both parties, same key does Encryption and decryption

. it is used for larger files.

Properties

-> it is used for large file Encryption
-> Has Same Key for both parties
-> AA.